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Halbach-Based Magnetic Configuration for Reduced Instrument Interference in Minimally Invasive Surgery

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***Abstract*— Lung cancer remains a leading cause of cancer mortality worldwide, and minimally invasive thoracic surgery has become the preferred approach for early-stage cases. Video-Assisted Thoracic Surgery (VATS) enables safe and oncologically effective resections through small incisions. To further reduce invasiveness, a magnetically anchored endoscope has been developed to share a single uniportal incision with surgical instruments. However, the proximity of ferromagnetic tools causes undesired magnetic interference, compromising anchoring stability and surgical safety. This study investigates optimized magnetic configurations to mitigate instrument–endoscope interference while maintaining sufficient anchoring and translational forces. Using finite element modeling (FEM), multiple magnet arrangements were analyzed, and a triangular Halbach pair configuration was identified to minimize stray field interactions with surgical instruments. The optimized design achieved stable anchoring with reduced lateral disturbance compared to conventional coaxial configurations. The magnetic interference was reduced by a maximum of 93% with the triangular prism array design, while maintaining the desired anchoring and translational performance. Experimental validation using fabricated magnets confirmed the simulation results and demonstrated effective interference suppression. The findings provide a practical magnetic design guideline for single-port magnetically anchored thoracoscopic systems, contributing to safer and more compact minimally invasive thoracic surgery.**

I. INTRODUCTION

Magnetic methods have proven effective in a wide range of medical applications [1], particularly in scenarios where the clinical workspace is confined [2], minimal incisions are desired [3], and targeted drug [4] or catheter delivery [5] to hard-to-reach regions is required. They have also shown great potential in endovascular interventions for disease treatment [6]. In particular, magnetically actuated surgical instruments have demonstrated unique advantages in manipulation tasks such as liver retraction during bariatric surgery [7], kidney stone retrieval [8], suturing [9], and tissue lifting for endoscopic submucosal dissection (ESD) [10]. These instruments exploit magnetic actuation to generate both attractive forces and control torques for precise manipulation.

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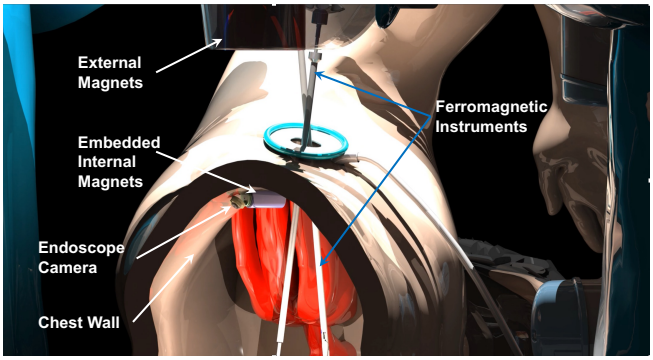


Fig. 1. Illustration of the surgical setup showing the external and internal permanent magnets, endoscopic camera, and ferromagnetic instruments in a single-port Video-Assisted Thoracic Surgery (VATS) scenario. The system employs a magnetically anchored, robotically manipulated intelligent endoscope for enhanced visualization and instrument coordination.

In such scenarios, internally deployed permanent-magnet-based devices operate in situ by clamping onto or penetrating tissue, organs, or calculi, and respond to the motion of the external magnetic field accordingly, with minimal magnetic interference from nearby surgical instruments.

However, in certain translation-based magnetic manipulation scenarios, the internal permanent-magnet-based device moves along anatomical structures in response to external magnetic locomotion. Examples include novel endoscopic control systems for minimally invasive laparoscopic surgery [11] and magnetically anchored endoscopes for Video-Assisted Thoracic Surgery (VATS) [12]. In these cases, undesired magnetic interference may occur when nearby surgical instruments are ferromagnetic and positioned too close to the internal magnet. Moreover, the presence of surgical instruments can also affect the accuracy of electromagnetic navigation systems [13]. The severity of such interference depends on the instrument's properties (e.g., powered electrosurgical, metallic, or ferromagnetic) and its proximity to the magnetic device. Experimental studies comparing instruments of different sizes, materials, and shapes have shown that stainless-steel tools should be kept at least 10 cm away to minimize magnetic navigation errors [14]. Further investigations have modeled and experimentally quantified interference caused by external electromagnetic coils acting on nearby internal permanent-magnet instruments that are not magnetically paired but are placed in confined abdominal spaces [15]. These studies aimed to determine practical separation distances that reduce interference while maintaining surgical feasibility. Other efforts have

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introduced soft-link mechanisms to position internal magnets away from the operation site, thereby avoiding interference with conventional laparoscopic tools while preserving stable tissue retraction [16]. Finally, expert consensus guidelines on magnetic anchoring and traction techniques in thoracoscopic and laparoscopic surgery recommend employing soft magnetic materials—such as pure iron, permalloy (Fe–Ni alloy), Fe–Co alloy, or silicon steel—as shielding layers around internal magnets to minimize interference with surrounding ferromagnetic instruments. This configuration effectively reduces magnetic flux leakage. Alternatively, a soft internal magnet can also be used to achieve comparable attenuation of stray magnetic fields [17].

While soft magnetic materials are effective for shielding and flux attenuation, achieving strong anchoring and actuation forces within confined anatomical spaces still requires high-performance permanent magnets. Among commercially available materials, neodymium–iron–boron (NdFeB) offers the highest magnetization and maximum energy product, thereby maximizing magnetic field strength per unit volume. For magnetic anchoring applications, rigid NdFeB-based permanent magnets therefore remain the optimal choice for compact and stable designs. At the same time, surgical systems are increasingly evolving toward robotic and automated operation, where robotic manipulators provide a precise and dexterous platform for carrying surgical instruments, including external magnetic sources for actuation. However, due to the limited payload capacity of robotic arms, optimization of the external magnetic configuration becomes essential to achieve sufficient field strength and gradient without exceeding mechanical constraints. To address this, optimized magnetic arrays have been developed to generate high magnetic field gradients—on the order of 7 T/m—producing strong attractive forces toward target regions [18]. Similarly, a flexible magnet array design has demonstrated an amplified and focused magnetic field approximately 1.8 times stronger than that of a cylindrical magnet of equal mass [19]. Despite these advancements, most reported studies focus on optimizing external magnet configurations, whereas internal magnet designs have rarely been explored using an array-based approach.

In this paper, we present a proof-of-concept investigation applying a Halbach-array magnet configuration [20] to the optimization of the internal magnet, aiming to minimize magnetic interference while preserving the desired anchoring and translational forces. The approach is implemented on a magnetically anchored and robotically navigated endoscope for Video-Assisted Thoracic Surgery (VATS), as shown in Figure 1. As illustrated, ferromagnetic instruments operating near the endoscope may interact with the tail region of the internal magnet embedded within the endoscopic camera, which is anchored to the chest wall by an externally manipulated magnet attached to a robotic arm. A finite element method (FEM)-based simulation study was conducted to compare multiple internal magnet configurations, including geometry-adjusted designs and two Halbach-array arrangements, while maintaining constant mass and dimensions

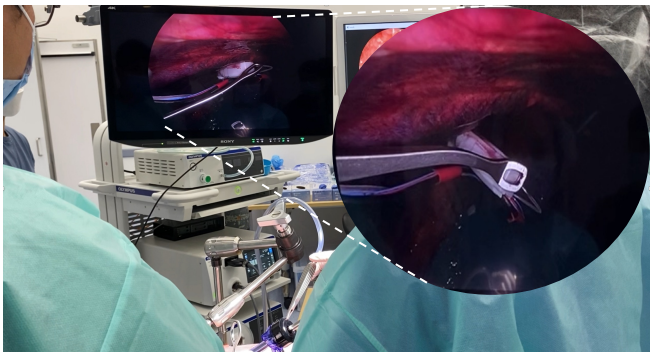


Fig. 2. Cadaver experiment showing detachment of the internal magnet from the chest wall due to magnetic interference from a nearby surgical instrument (SCANLAN® D'Amico Biopsy Forcep, SCANLAN INTERNATIONAL, Inc., USA). The test was conducted using the initial version of the magnetically anchored and guided endoscope system for single-port VATS.

for both the external and internal magnets. Based on the simulation results, the best-performing configuration was fabricated and evaluated in a proof-of-concept experimental setup, validating the feasibility and effectiveness of the proposed design.

The remainder of this paper is organized as follows. Section II presents the force and torque constraints and identifies the clinical instrument interference problem of the initial magnetic anchoring and navigation system. Section III introduces the methods and magnetic configurations compared in this study. Section IV presents the simulation results and the proof-of-concept experimental validation, where the optimized magnet configuration was customized for testing. Finally, Section V concludes the paper.

II. SYSTEM SETUP

The system is composed of an external magnetic source, held by a manual or robotic arm, that generates translational and rotational motion, and an internal magnetic response unit embedded in a customized thoracic endoscope. The internal unit is anchored to the chest wall and achieves locomotion through alignment with the external magnetic source. Given the confined space near the single port, NdFeB permanent magnets were selected for both the external and internal magnetic units. During surgery, the magnetic endoscope is first inserted into the chest cavity through the single port, anchors onto the chest wall to provide visual feedback, and subsequently allows surgical instruments to reuse the same port for operation, as shown in Figure 1.

A. System Constraints

Without considering any external interference, to ensure that the endoscope is operated properly during the thoracic surgery, the fundamental magnetic anchoring safety condition must be satisfied as a premise. As illustrated in Figure 3a and b, the internal and external magnets are diametrally magnetized, with their centers aligned along the z-axis and separated by a distance d. The internal magnet experiences both the magnetic attraction from the external magnet and the gravitational force acting downward.

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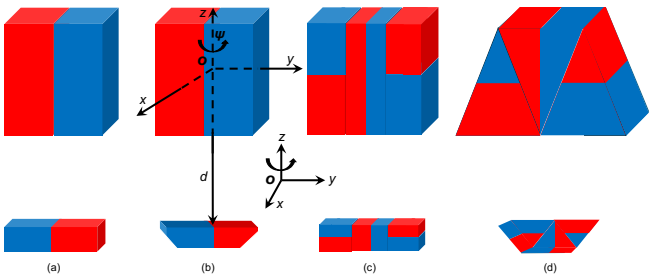


Fig. 3. Illustration of the explored magnetic configurations used for manipulation of a magnetically anchored endoscope designed for chest wall anchoring and navigation in single-port Video-Assisted Thoracic Surgery (VATS).

Simplified diametral dipole model.: Each magnet is modeled as a transverse magnetic dipole: $\mathbf{m}_1 = m_1\,\hat{\mathbf{x}}$ for the external magnet and $\mathbf{m}_2 = m_2(\cos\psi\,\hat{\mathbf{x}} + \sin\psi\,\hat{\mathbf{y}})$ for the internal magnet, where ψ is the in-plane angle between their magnetization directions. The center-to-center separation is along $+\hat{\mathbf{z}}$, and gravity acts along $-\hat{\mathbf{z}}$.

Axial magnetic force.: Under the diametral dipole approximation, the axial magnetic force acting on the internal magnet is given by

$$F_{\mathrm{mag},z}(d) \;=\; \frac{3\mu_0}{4\pi}\,\frac{m_1 m_2\cos\psi}{d^4}, \tag{1}$$

where μ_0 is the vacuum permeability, and ψ determines the effective coupling between the two dipoles. A positive $\cos\psi$ corresponds to attraction (upward pull), while a negative value represents repulsion.

Torque alignment.: The external field also induces a restoring torque on the internal magnet:

$$\tau_z(d) \;=\; \frac{\mu_0}{4\pi}\,\frac{m_1 m_2}{d^3}\,\sin\psi, \tag{2}$$

which tends to align the internal magnet's diametral magnetization with that of the external magnet ($\psi \to 0$).

Magnetic anchoring safety condition.: To maintain magnetic anchoring against gravity, the net vertical force on the internal magnet must remain positive:

$$F_{\mathrm{mag},z}(d) - mg \;>\; 0, \tag{3}$$

where m is the mass of the combined internal robot (internal magnet and the associated endoscope) and g is the gravitational acceleration. Equation (3) defines the basic safety constraint for stable magnetic anchoring during thoracic procedures. The combined internal robot is expected to be stably anchored on the chest wall, as illustrated in Figure 1.

Here, $\mathbf{m}_2$ denotes an equivalent dipole representing the internal magnet, rather than a specific Halbach element or array. This simplified model is used to explain the anchoring and interference mechanisms, while FEM is later employed to evaluate realistic Halbach-array geometries.

B. Interference Problem

With the initially implemented magnetic configuration (illustrated in Figure 3a), severe magnetic interference between the ferromagnetic surgical instruments and the magnetically anchored endoscope for single-port Video-Assisted Thoracic Surgery (VATS) was identified during an earlier cadaver experiment, as shown in Figure 2. During the surgical procedure, when a ferromagnetic instrument approached the internal magnet, a strong magnetic attraction force was induced between them, introducing an additional variable that compromised anchoring stability. Further tests demonstrated that all commonly used surgical instruments for thoracic procedures are ferromagnetic and exhibit strong attraction to the internal magnet, often leading to anchoring failure.

To ensure the endoscope remains stably anchored during the procedure, the overall magnetic equilibrium must now account for the additional attraction between the internal magnet and the nearby ferromagnetic instrument. Let $F_{\mathrm{inst}}(d_i)$ denote the magnetic attraction force between the internal magnet and the approaching ferromagnetic instrument, where d_i represents their instantaneous separation distance, and $F_{\mathrm{mag},z}(d)$ is the anchoring force between the external and internal magnets as defined in Equation (1). The net vertical force acting on the internal magnet can thus be expressed as

$$F_{\mathrm{net}}(d, d_i) \;=\; F_{\mathrm{mag},z}(d) \;-\; F_{\mathrm{inst}}(d_i) \;-\; mg, \tag{4}$$

where the second term represents the downward disturbance force generated by the instrument–magnet attraction.

Updated magnetic anchoring safety condition.: For the internal magnet to maintain stable anchoring against both gravity and the induced disturbance, the net vertical force must remain positive:

$$F_{\mathrm{net}}(d, d_i) \;=\; F_{\mathrm{mag},z}(d) \;-\; F_{\mathrm{inst}}(d_i) \;-\; mg \;>\; 0. \tag{5}$$

When the ferromagnetic instrument approaches the internal magnet ($d_i \downarrow$), the term $F_{\mathrm{inst}}(d_i)$ increases rapidly due to the nonlinear dependence on distance, reducing the net upward anchoring force. If $F_{\mathrm{inst}}(d_i)$ exceeds $F_{\mathrm{mag},z}(d) - mg$, the anchoring force in Equation (5) becomes negative, resulting in detachment or sudden displacement of the internal magnet.

Remarks.: (i) The disturbance force $F_{\mathrm{inst}}(d_i)$ can be modeled empirically or approximated using a magnetic dipole–soft material interaction model, e.g., $F_{\mathrm{inst}}(d_i) \propto \mu_0 M_{\mathrm{int}} M_{\mathrm{ferro}}/d_i^n$ with $n \in [2, 4]$ depending on geometry. (ii) Equation (5) therefore defines the operational constraint for magnetic anchoring safety during single-port VATS, imposing a minimum allowable separation between the ferromagnetic instruments and the internal magnet.

Clinical Reality.: In practice, frequent exchange and manipulation of instruments such as graspers, retractors, and forceps through a single uniportal access are unavoidable. During single-port Video-Assisted Thoracic Surgery (VATS), surgeons must alternate among these tools to perform sequential tasks including tissue retraction, vessel isolation, and dissection within a confined operative space. Because all instruments pass through the same incision, their shafts and end-effectors inevitably approach or momentarily contact the internal magnet used for endoscopic anchoring. These repeated unpredictable interactions induce transient disturbance forces $F_{\mathrm{inst}}(d_i)$, as described above, which can perturb the local magnetic field and compromise anchoring stability.

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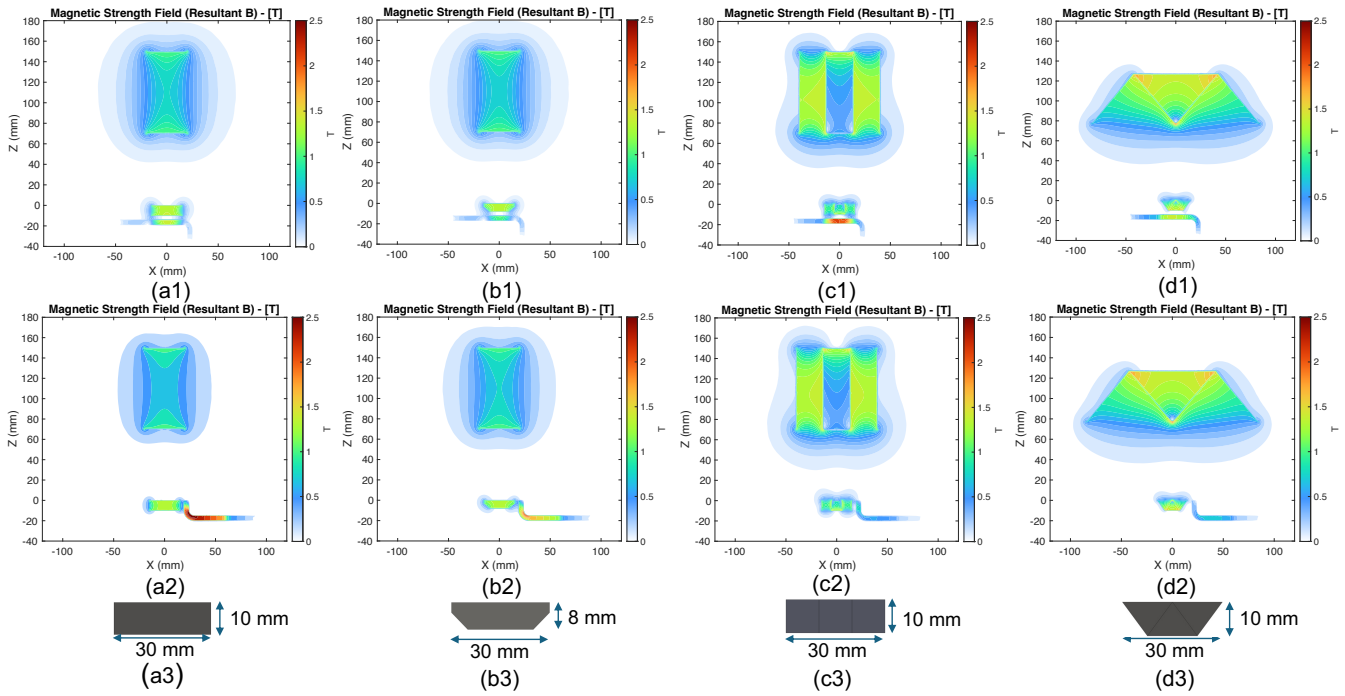


Fig. 4. FEM analysis of magnetic interference caused by a simulated ferromagnetic instrument for the explored magnetic configurations shown in Figure 3. The configurations are designed to resist interference when a ferromagnetic instrument approaches the bottom or tail region of the internal magnet, ensuring stable chest wall anchoring and navigation of a magnetically anchored endoscope in uniportal Video-Assisted Thoracic Surgery (VATS).

Most thoracoscopic instruments—such as dissectors, scissors, suction probes, and staplers—are made of ferromagnetic stainless steels (AISI 420/440 or 17-4 PH), exhibiting strong attraction to the internal magnet. Although non-magnetic alternatives (e.g., titanium or austenitic stainless steel 316L) exist, they are seldom used in standard VATS sets. This reality underscores the need to design and implement a **contact-safe** internal magnet configuration that mitigates magnetic interference and preserves anchoring safety during instrument exchange.

III. METHODS

In this section, several potential magnetic configurations are presented to minimize interference caused by ferromagnetic surgical instruments. The finite element method (FEM) setup with a simplified ferromagnetic instrument, and the mass and geometric constraints are also detailed.

A. The Proposed Magnetic Configurations

The initial cadaver experiment (Figure 2) revealed that the rectangular internal magnet used in the first prototype (Figure 3a) resulted in severe magnetic interference when it contacted ferromagnetic instruments near its bottom or tail region. To address this issue, a shape-optimized internal magnet was proposed as a potential solution. As shown in Figure 3b, the width of the internal magnet was reduced, and both rear corners were trimmed into triangular edges to minimize the contact area with the instruments.

In addition, arranging an array of permanent magnets can augment the magnetic field on one side of the array while nearly canceling it on the opposite side, achieved through a spatially rotating pattern of magnetization. Therefore, further optimizations were explored by adapting the Halbach magnet array into the magnet pair, as shown in Figure 3c and Figure 3d, composed of rectangular cuboid and triangular prism units, respectively.

B. FEM Setup for Comparison

To quantify the magnetic interference force, a simplified L-shaped ferromagnetic instrument was introduced into the FEM simulation. Its dimensions and material properties (permeability and conductivity) were kept identical across all four magnetic configurations shown in Figure 3. The ferromagnetic instrument was positioned beneath the internal magnet to evaluate its influence on endoscopic anchoring, as shown in Figure 4a1–d1.

In the simulation, the anchoring distance d between the internal and external magnets was kept constant at 70 mm, while the vertical gap between the top surface of the instrument and the underside of the internal magnet was varied from 6 mm to 16 mm. To evaluate interference when contact occurred at the tail surface, the ferromagnetic instrument was placed on the right side of the internal magnet (Figure 4a2–d2). During this simulation, the horizontal separation between the instrument and the right edge of the internal magnet was varied from 6 mm to 16 mm, while maintaining the same anchoring distance of 70 mm. The anchoring distance was set to 70 mm to account for extreme clinical scenarios, including increased chest wall thickness in obese patients.

For fair comparison, the mass of the external magnet was constrained to 1.6 kg, limited by the payload capacity of

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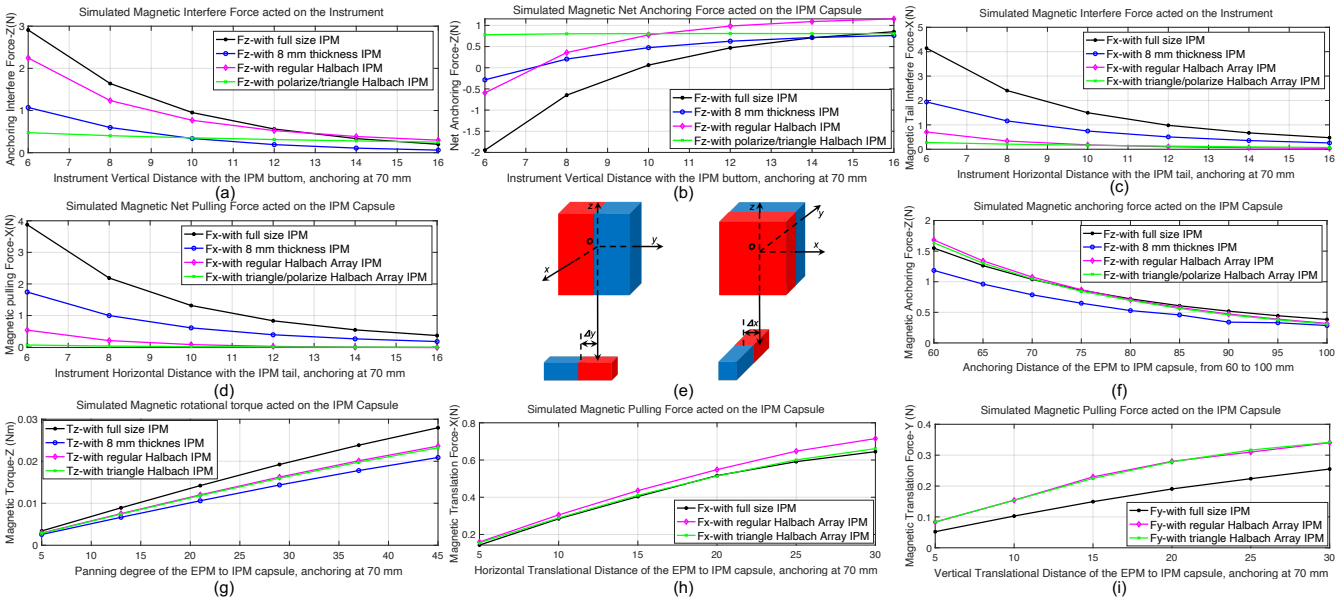


Fig. 5. Plots (a)–(d) present the FEM analysis results of magnetic interference caused by a simulated ferromagnetic instrument for the explored magnetic configurations shown in Figure 3. Plot (e) illustrates the definition of the translational offset caused by the external magnets. Plots (f)–(i) show that the anchoring force, panning torque, and translational force performance of the triangular Halbach magnetic configuration are comparable to, or even better than, those of the baseline configuration.

the magnet holder. The dimensions of the internal magnet were restricted to 30 mm × 10 mm × 10 mm (length, width, and thickness) to accommodate the single-port access constraints. All magnets were made of NdFeB material magnetized to grade N52. Assuming a density of 7,500 kg/m^3, the external magnets were configured as follows: a single 47.4 × 47.4 × 94.8 mm block, triple 94.8 × 23.7 × 94.8 mm blocks, and triple 50 × 45 × 63 mm blocks, respectively.

IV. SIMULATION RESULTS AND PROOF-OF-CONCEPT EXPERIMENTS

Based on the proposed magnetic configurations and the experimental environment setup, a comprehensive simulation study was conducted, from which the best-performing configuration—the triangular Halbach approach—was selected. The structural safety of the assembly was analyzed, and the magnets were fabricated for validation. Proof-of-concept experiments confirmed that the selected configuration effectively minimizes magnetic interference while maintaining the desired anchoring, translational, and rotational capabilities.

A. Simulation Results

The bottom-side interference was first analyzed, and the results are shown in Figure 5a–b. It can be observed that the magnetic force acting on the surgical instrument and the net anchoring force on the internal magnet vary significantly, with the original configuration exhibiting the highest level of interference. While the corner-trimmed configuration reduces disturbance due to additional separation (2 mm thinner), and the rectangular-cuboid-based Halbach array further decreases interference, only the triangular-prism-based array resulted in a positive net anchoring force. The interference at a separation distance of 6 mm was reduced by approximately 83% for the triangular-prism configuration compared to the original configuration. Furthermore, the results for interference caused by the rear side are shown in Figure 5c–d. The results indicate that corner removal again reduced the undesired pulling force, while both Halbach array designs significantly minimized the disturbance, with the triangular-prism-based array yielding the lowest interference. This makes the triangular-prism-based configuration the preferred option. At a separation distance of 6 mm, the triangular-prism configuration reduced rear-side interference by approximately 93% compared to the original configuration.

To further confirm the suitability of the configuration using the triangular-prism array, the desired anchoring, translational, and rotational capabilities were examined in the absence of a ferromagnetic instrument. The definitions of anchoring and panning are provided in Figure 3b, and translation is defined in Figure 5e. The results shown in Figure 5f–h confirm that the triangular-prism configuration maintains effective anchoring, panning, and translation along the y-direction, while its translation along the x-direction is predicted to be even better than that of the original configuration, as shown in Figure 5i.

B. Experimental Validation

The triangular-prism-array configuration was selected for further experimental validation. As shown in Figure 6a, the internal magnetic forces of the selected array were examined and confirmed that the array was inherently self-assembled without the need for an external holding structure. It is worth noting that, in order to accommodate the endoscope camera cable and rinsing tube, a working channel with a diameter of 3 mm was reserved at the center of the internal triangle magnet unit, as shown in Figure 6a1. The fabricated and

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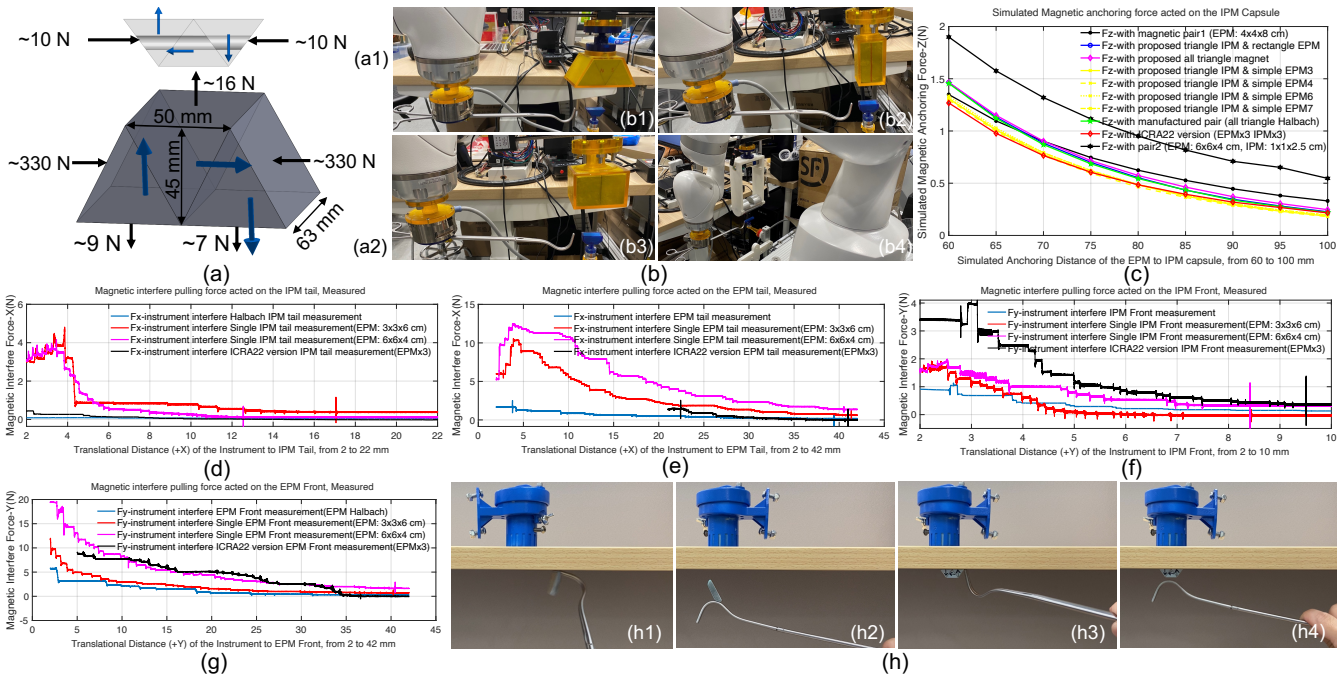


Fig. 6. Experimental results with the fabricated triangular Halbach magnet configuration. (a) Internal magnetic force and dimensions of the triangular prism array. (b) Experimental setup comparing the newly fabricated configuration with the existing designs. (c) Anchoring force comparison among configurations. (d)–(g) Interference force measurements on both the external and internal magnets. (h) Experimental validation with freely movable internal magnets. A supplementary video (`supplementary-video-1.mp4`) demonstrating magnetic interference caused by commonly used ferromagnetic instruments (SCANLAN® Harken Clamps, SCANLAN INTERNATIONAL, Inc., USA) in VATS is provided, validating that the triangular Halbach configuration significantly reduces interference, resulting in stable and safe endoscopic anchoring even under instrument–magnet contact.

assembled magnetic pair, along with the previously customized magnet pairs, ferromagnetic instrument, and force measurement setup, are shown in Figure 6b1–b4.

The anchoring force induced by these fabricated magnet pairs was simulated and plotted in Figure 6c, reflecting the actual manufactured dimension. It can be observed that the anchoring force of the fabricated Halbach triangular prism array is smaller than the simulation results in Figure 5f, due to the reserved working channel. However, it can still generate sufficient anchoring force, which is higher than that of a previously developed six-degree-of-freedom version [12], as shown in Figure 6b4. The magnetic interference forces induced by the ferromagnetic instrument, acting on the front (bottom-replaced due to the internal mount constraint) and tail sides of the internal and external magnets, were measured and presented in Figure 6d–g. It is demonstrated that an additional magnet pair (EPM: 60 x 60 x 60 mm, IPM: 10 x 10 x 25 mm), modified from the configuration shown in Figure 3b - with a larger external magnet and shorter internal magnet - can generate a much higher anchoring force, as shown in Figure 6c. However, it is still severely affected by the interference from the instrument, with only minor improvement observed when the instrument approaches the tail. These results validated the superior anti-interference capability of the triangular-prism-based Halbach array when the ferromagnetic instrument operated nearby.

Finally, experiments conducted under a free-movement condition mimicking the VATS scenario compared the corner-trimmed thinner magnet with the triangular-prism magnet array, demonstrating that the proposed design effectively minimized magnetic disturbance while maintaining stable anchoring. This advantage arises primarily because the triangular prism geometry outperforms rectangular Halbach arrays by enabling superior field cancellation at slanted interfaces—where opposing radial and axial magnetization components cancel locally—resulting in reduced stray fields and weak-side interference, with effective distance playing only a secondary role.

V. CONCLUSION

In this study, we identified and analyzed the magnetic interference problem caused by the contact between ferromagnetic surgical instruments and the internal magnet, which limits the broader application of magnetic actuation techniques in minimally invasive laparoscopic and thoracic surgery. Several alternative configurations were proposed, and FEM simulations were conducted to evaluate clinical scenarios in which ferromagnetic instruments approached the internal magnet. Simulation results indicated that the triangular-prism Halbach array provided the best performance, reducing magnetic interference by up to 93% while maintaining stable anchoring and translational capability. Experiments using a fabricated prototype further validated the design in a proof-of-concept setup. The findings of this work provide a practical guideline for the design of contact-safe magnetic anchoring systems and are expected to benefit future magnetic-based surgical robotic platforms requiring compact and interference-robust actuation mechanisms.

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